Lab 6: Derived Variables and Linear Regression

In this session, you will use data from the 2019 National Survey of Early Care and Education (NSECE) Workforce Questionnaire to practice data management, create derived variables, evaluate the assumptions of Ordinary Least Squares (OLS) regression, and estimate linear regression models.

The goal of this exercise is to examine whether participation in professional development activities is associated with higher hourly wages among center-based early care and education staff in the United States.

1. Set up your working environment

Load the following packages: tidyverse, ggplot2.

2. Prepare the data

Subset the NSECE dataset to retain the following variables:

- WF9_WORK_FT – Staff full-time or part-time employment status
- WF9_CHAR_GENDER – Staff gender
- WF9_CAREER_EXPERIENCE – Years of experience caring for children under age 13
- WF9_CHAR_EDUC – Highest level of education completed
- WF9_CHAR_RACE – Staff race
- Professional development activities: WF9_PROFDEV_WRKSHP, WF9_PROFDEV_COACH, WF9_PROFDEV_MEETING, WF9_PROFDEV_COURSE
- WF9_WORK_WAGE – Hourly wage

Rename each variable using clear, descriptive names. For each variable, identify its measurement level (nominal, ordinal, or interval/ratio) and indicate whether it is discrete or continuous.

Next, create the following derived variables:

- Full-time: 1 if the staff member worked full time; 0 otherwise.
- Female: 1 if the staff member identified as female; 0 otherwise.
- Create k-1 binary variables for years of experience working with children (k = number of categories). After examining the variable, choose a reference group.
- Create k-1 binary variables for education level (k = number of categories). After examining the variable, choose a reference group.
- Create k-1 binary variables for race (k = number of categories). After examining the variable, choose a reference group.
- Professional development: 1 if the staff member answered "Yes" to any of the following activities WF9_PROFDEV_WRKSHP, WF9_PROFDEV_COACH,

WF9_PROFDEV_MEETING, WF9_PROFDEV_COURSE. Code the variable as 0 otherwise.

3. Linear Regression

Previous research suggests that staff who participate in professional development activities may earn higher hourly wages than those who do not.
To examine this relationship, estimate the following OLS regression model.

$$\text{Hourly Wage} = \beta_0 + \beta_1(\text{Full Time}) + \beta_2(\text{Female}) + \beta_{3\ldots}\beta_7(\text{Years of Experience categories}) + \beta_{8\ldots}\beta_{13}(\text{Education Level categories}) + \beta_{14\ldots}\beta_{16}(\text{Race categories}) + \beta_{17}(\text{Professional Development}) + \varepsilon$$

4. OLS assumptions and interpretation

Create the residual plot and a histogram of the model residuals.

Finally, answer the following questions:

1. Do the residual plots suggest that the assumptions of linearity and homoscedasticity are reasonably met?
2. Does the histogram indicate that the residuals are approximately normally distributed?
3. Interpret the coefficient estimates for each independent variable.
4. Interpret the overall model fit (e.g., R^2 and adjusted R^2).
5. Based on your findings, what conclusions can you draw about the relationship between professional development and hourly wages among center-based early care and education staff?